

Beyond Calibration: Regime-Dependent Car-Following of Autonomous versus Human-Driven Vehicles and Its Emergent String-Stability and Safety Consequences

Pedram Beigi, Mohammad Emad Rashidi, Nachuan Li, Shayan Bafandkar,
Dana Monzer, John Hourdos, Hani Mahmassani, Alireza Talebpour, Samer H. Hamdar

Abstract

This study moves beyond descriptive parameter comparison to examine how the longitudinal behavior of small vehicles, large vehicles, and autonomous vehicles (AVs) translates into emergent traffic-flow and safety outcomes. Using high-resolution TGSIM trajectory data, we calibrate two representative car-following models—the physics-based Intelligent Driver Model (IDM) and a behaviorally motivated Prospect-Theory (PT) model—at the individual leader-follower level using a *closed-loop* trajectory simulation, in which the spacing that drives the model is recomputed from the simulated follower position at every step. We make three contributions. First, we stratify car-following by traffic regime (free-flow, synchronized, and stop-and-go) and show that class differences which are diluted in the pooled analysis emerge within specific regimes. Second, we assess *out-of-sample* predictive accuracy and *practical identifiability*: using the dispersion of multi-start calibration and one-at-a-time profiles, we show that IDM parameters related to spacing and desired speed are well identified and transferable, whereas several PT parameters are practically non-identifiable from short trajectories—reframing the limited PT differentiation as an identifiability property rather than an absence of behavioral difference. Third, we map the calibrated class- and regime-specific parameters to emergent dynamics through an analytical linear string-stability criterion and mixed-autonomy platoon simulations, quantifying string amplification, surrogate safety measures (time-to-collision exposure and required deceleration), and throughput across AV market-penetration rates. The results show that the calibrated AV behavior trades modest throughput gains against reduced string stability and shrinking safety margins at high penetration, with consequences that are strongly regime dependent. The framework yields class- and regime-specific parameters that are directly usable as priors for mixed-autonomy simulation and for lane-management and AV-integration policy.

Keywords: Car-following, Autonomous Vehicles, Prospect Theory, Traffic Regimes, String Stability, Parameter Identifiability, Naturalistic Trajectory Data.

1 Introduction

Microscopic simulation is a pivotal tool for the analysis and design of traffic systems, allowing controlled study of new technologies without disrupting real traffic [?]. Car-following (CF) models are the core of microscopic simulation, reproducing the longitudinal motion of a vehicle behind its leader [?]. Despite decades of development, our understanding of CF behavior remains incomplete, constrained by data, by the difficulty of model calibration and validation, and—most acutely—by the heterogeneity of driving agents, which now includes autonomous vehicles (AVs) [?].

AVs are expected to increase throughput and improve stability, but the empirical evidence is mixed: commercial adaptive-cruise-control systems have been shown to be string *unstable* in several field experiments [?]. While AV longitudinal control has been studied extensively, there remains limited empirical clarity on (i) how AV behavior differs from human-driven vehicles (HDVs) *across traffic regimes*, (ii) whether

the calibrated parameters are *identifiable and transferable*, and (iii) what these parameters imply for *emergent* flow and safety in mixed traffic. The first version of this study calibrated IDM and PT per leader–follower pair and compared the resulting parameters statistically. That analysis, while internally consistent, shared the limitations of much of the literature: it pooled all traffic conditions, reported only in-sample fit, and stopped at the parameters themselves without connecting them to emergent behavior.

This paper addresses those gaps directly. We use the naturalistic TGSIM dataset [?] and classify the observed followers into (i) *small* vehicles (passenger cars, SUVs, pickups), (ii) *large* vehicles (trucks and buses), and (iii) *autonomous* vehicles. Our contributions are:

1. **Regime-stratified calibration and comparison.** Each car-following episode is labeled by traffic regime (free-flow, synchronized, stop-and-go), and class differences are tested both pooled and within regime, revealing regime-specific behavioral distinctions that pooling conceals.
2. **Validation and practical identifiability.** We evaluate out-of-sample prediction (calibrate on part of an episode, predict the rest) and quantify practical identifiability from multi-start dispersion and profile analysis, establishing which parameters are trustworthy and explaining the limited PT differentiation as non-identifiability.
3. **Emergent string-stability and safety analysis.** We translate the calibrated class/regime parameters into analytical linear string stability and mixed-autonomy platoon simulations, quantifying string amplification, surrogate safety, and throughput as functions of AV market penetration.

2 Literature Review

2.1 Calibration of car-following models

Among physics-based models, the IDM [?] reproduces realistic fundamental diagrams and stable/unstable regions and consistently shows low calibration error relative to alternatives [?, ?]. Trajectory-based calibration is now standard [?], but a recurrent methodological pitfall is the choice of simulation scheme. In an *open-loop* (one-step) scheme the model is fed observed states at each step; in a *closed-loop* scheme the spacing and speed are integrated forward and the spacing that drives the model is recomputed from the simulated position. Punzo and co-authors emphasize that closed-loop, spacing-based objectives are required for trajectory calibration to be sensitive to the parameters and to avoid optimistic, non-identifiable fits [?, ?]. We adopt a closed-loop scheme for *both* models in this study.

2.2 Behavioral models and Prospect Theory

Psychologically grounded models capture driver decision-making under uncertainty. The Prospect-Theory CF model [?, ?] represents acceleration as a risk-taking choice combining subjective utilities for accelerations and disutilities for crash risk, calibrated with a genetic algorithm to account for inter-driver heterogeneity. PT-based formulations have been extended to AV–HDV mixed traffic [?, ?] and to connected-vehicle compliance [?]. A known difficulty with rich behavioral models is parameter identifiability: many parameters interact and may not be separately recoverable from short trajectories [?].

2.3 AV car-following and string stability

Several studies characterize adaptive-cruise-control and AV CF behavior using field data [?, ?], with conflicting conclusions on which model best fits AV trajectories [?, ?]. Hu et al. [?] calibrate IDM on the Waymo dataset and conclude AVs are more conservative (longer headways, larger jam spacing) and less prone to

string instability, while Alhariqi et al. [?] introduce an adaptive IDM for mixed autonomy. String stability—the (non-)amplification of speed perturbations along a platoon—is a classical property of CF models [?, ?], yet it is rarely connected back to *empirically calibrated* class-specific parameters and almost never stratified by traffic regime. This paper makes that connection explicit.

3 Methodology

3.1 Data

We use the naturalistic Third-Generation Simulation (TGSIM) dataset [?], which captures automated and human-driven vehicles across several U.S. facilities. We use the I-395 (Washington, D.C.) urban expressway segment and the I-294 (Illinois) segments. A valid car-following episode requires: duration > 10 s; bumper-to-bumper gap < 200 m and > 0 ; no lane change; and a constant preceding-vehicle identity throughout. Followers are labeled small, large, or autonomous. To ensure comparability, equal numbers of small and large followers are sampled in the spatial–temporal neighborhood of each AV episode.

3.2 Car-following models

Intelligent Driver Model (IDM). The acceleration is

$$\dot{v}_n(t) = a \left[1 - \left(\frac{v_n(t)}{v_0} \right)^\delta - \left(\frac{s^*(v_n, \Delta v_n)}{s_n(t)} \right)^2 \right], \quad (1)$$

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$$s^*(v_n, \Delta v_n) = s_0 + \max \left(0, v_n T + \frac{v_n \Delta v_n}{2\sqrt{ab}} \right), \quad (2)$$

with parameters $\Theta_{\text{IDM}} = \{T, a, b, v_0, s_0\}$, where T is the safe time headway, a the maximum acceleration, b the comfortable deceleration, v_0 the desired speed, and s_0 the minimum bumper-to-bumper gap. The acceleration exponent δ is fixed at the standard value $\delta = 4$; as shown in Section ??, δ is not practically identifiable from short trajectories and “calibrating” it injects noise.

Prospect-Theory (PT) model. The PT model [?, ?] selects acceleration as a risk-taking decision. The generalized utility combines a prospect-theoretic acceleration utility and a crash-risk penalty,

$$U(a; s, v, \Delta v) = U_{\text{PT}}(a) + U_{\text{crash}}(a; s, v, \Delta v), \quad (3)$$

and the realized acceleration is obtained by a Newton–Raphson solution of the first-order condition (full equations as in [?]). The calibrated parameters are $\Theta_{\text{PT}} = \{T_{\text{max}}, \alpha, \beta, W_c, W_m, \gamma\}$: the maximum anticipation horizon T_{max} , speed-uncertainty coefficient α , intra-driver logit variability β , accident weight W_c , negative-utility asymmetry W_m , and sensitivity exponent γ .

3.3 Closed-loop trajectory simulation

For both models the follower trajectory is integrated forward from the first observed state $(x_n(0), v_n(0))$ using the observed leader trajectory. Crucially, the spacing that enters the model is recomputed at each step from the *simulated* follower position,

$$s_n(t_i) = x_{\text{lead}}(t_i) - x_n(t_i) - \frac{1}{2}L_{\text{lead}} - \frac{1}{2}L_{\text{foll}}, \quad (4)$$

with a ballistic update $v_{i+1} = \max(0, v_i + a_i \Delta t)$, $x_{i+1} = x_i + v_{i+1} \Delta t$ (PT uses $x_{i+1} = x_i + v_i \Delta t + \frac{1}{2} a_i \Delta t^2$). This closed-loop scheme makes the simulated trajectory—and hence the calibration objective—genuinely sensitive to the parameters, in contrast to an open-loop scheme that feeds the observed spacing back into the model and yields optimistic, weakly identifiable fits. Acceleration is bounded to $[-8, 5]$ m/s².

3.4 Genetic-algorithm calibration

Parameters are calibrated per leader–follower episode by a genetic algorithm that minimizes a weighted trajectory error,

$$f(\Theta) = \sum_j \left(\lambda_p |x_{\text{obs},j} - x_{\text{sim},j}(\Theta)| + \lambda_v |v_{\text{obs},j} - v_{\text{sim},j}(\Theta)| \right), \quad (5)$$

with $\lambda_p = 1$, $\lambda_v = [0.5]$. To assess robustness and identifiability (Section ??) each episode is calibrated from $N = [20]$ random restarts.

3.5 Traffic-regime classification

Each episode is assigned a dominant traffic regime from its follower speed profile $v_n(\cdot)$ using interpretable features: mean speed $\bar{v}$, speed standard deviation σ_v , minimum speed $v_{\min}$, the near-stop fraction $\phi = \Pr(v < v_{\text{stop}})$, and the speed range $v_{\max} - v_{\min}$. With $v_{\text{stop}} = 3$ m/s, the rules are:

- **stop-and-go**: $\phi > 0.05$, or ($v_{\min} < v_{\text{stop}}$ and a large, oscillatory speed range);
- **free-flow**: $\bar{v} \geq 17$ m/s and $\sigma_v \leq 1.5$ m/s;
- **synchronized**: otherwise (congested but flowing).

A sliding window labels each instant and contiguous same-regime segments (≥ 10 s) define the dominant regime. Class comparisons are then performed both pooled and *within* each regime.

3.6 Out-of-sample validation and practical identifiability

Out-of-sample prediction. For each episode we calibrate on the first 70% of the trajectory and predict the remaining 30% in closed loop, reporting in-sample and out-of-sample position RMSE and their ratio. This guards against overfitting that pure in-sample error cannot detect.

Practical identifiability. For a parameter θ with restart estimates $\{\hat{\theta}^{(r)}\}_{r=1}^N$, the normalized restart dispersion

$$\text{CV}(\theta) = \frac{\text{std}_r \hat{\theta}^{(r)}}{|\hat{\theta}|} \quad (6)$$

measures how tightly the data pin down θ . We classify $\text{CV} \leq 0.10$ as well identifiable, $0.10 < \text{CV} \leq 0.30$ as moderately identifiable, and $\text{CV} > 0.30$ as poorly identifiable. We complement this with one-at-a-time profiles of the fit around the optimum: a flat profile indicates a non-identifiable direction.

3.7 Statistical comparison

Class differences are tested with Welch’s ANOVA (robust to unequal variances) and Games–Howell post-hoc pairwise tests (robust to unequal variances and sample sizes), reporting *F*/*t* statistics, *p*-values, and Hedges’ *g* effect sizes. The same tests are applied within each regime.

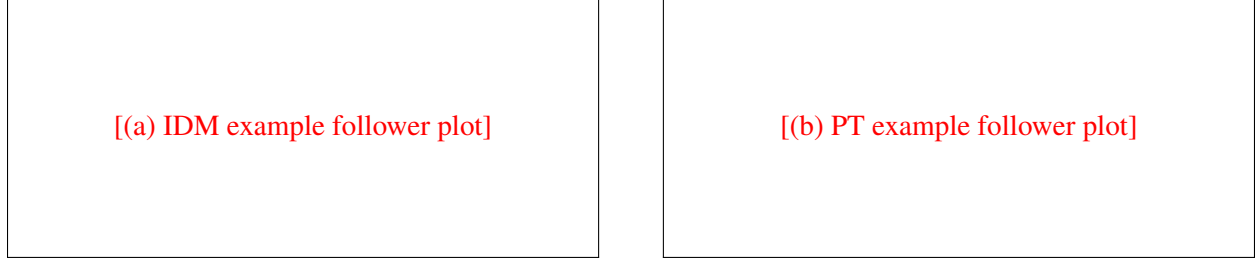


Figure 1: Closed-loop simulated (dashed) versus observed (solid) position and speed for a representative follower: (a) IDM, (b) PT.

3.8 Emergent dynamics

Linear string stability. Linearizing a CF model $\dot{v}_n = f(s_n, \Delta v_n, v_n)$ about a uniform equilibrium (v_e, s_e) (with $\Delta v = v_n - v_{n-1}$) gives the speed-to-speed transfer function between consecutive vehicles

$$H(\omega) = \frac{f_s - i\omega f_{\Delta v}}{(f_s - \omega^2) - i\omega(f_{\Delta v} + f_v)}, \quad (7)$$

where $f_s, f_v, f_{\Delta v}$ are partial derivatives of f at equilibrium. The platoon is string stable iff $|H(\omega)| \leq 1$ for all ω ; equivalently the low-frequency criterion

$$\lambda_2 = \frac{1}{2}f_v^2 + f_{\Delta v}f_v - f_s \geq 0 \quad (8)$$

must hold [?]. For IDM the equilibrium spacing is $s_e = (s_0 + v_e T) / \sqrt{1 - (v_e/v_0)^\delta}$ and partials are evaluated numerically. We report the peak gain $\max_\omega |H(\omega)|$ and the unstable speed band per class and regime.

Mixed-autonomy platoon simulation. Using class-specific calibrated parameters, we simulate a single-lane platoon at AV market-penetration rate (MPR) $p \in [0, 1]$ subject to a stop-and-go lead-vehicle perturbation, averaging over random class compositions. We report: the empirical string amplification (ratio of last-to-lead speed-oscillation standard deviation); surrogate safety measures time-to-collision $TTC = s/\Delta v$ (when closing), required deceleration $DRAC = \Delta v^2/(2s)$, time-exposed TTC below 3 s (TET) and time-integrated TTC (TIT); and throughput.

4 Results

4.1 Calibration accuracy and out-of-sample validation

Figure ?? shows simulated versus observed position and speed for representative IDM and PT followers under the closed-loop scheme. Table ?? reports mean RMSE and MAE by class for both models, together with the out-of-sample errors from the 70/30 temporal split. [Fill Table ?? from the re-run. Note that closed-loop RMSE will be larger than the previously reported open-loop values; this is the correct, honest metric. Report the in-/out-of-sample ratio to show the models generalize.]

4.2 Practical identifiability

Table ?? reports the median normalized restart dispersion (Eq. ??) per parameter. [Fill from analysis/validation_identifiability] We expect IDM's T , v_0 , and (in congested regimes) s_0 to be well identified, while several PT parameters ($T_{\max}$, γ , β) are poorly identified. This reframes the limited PT differentiation observed across classes: where

Table 1: Trajectory error by vehicle class (closed-loop), in-sample and out-of-sample.

Model	Vehicle type	RMSE (m)	MAE (m)	OOS RMSE (m)	OOS/IS ratio
IDM	Small	□	□	□	□
	Large	□	□	□	□
	AV	□	□	□	□
PT	Small	□	□	□	□
	Large	□	□	□	□
	AV	□	□	□	□

Table 2: Practical identifiability: median restart-dispersion CV per parameter (lower = better identified).

Parameter	Small	Large	AV
IDM T	□	□	□
IDM a	□	□	□
IDM b	□	□	□
IDM v_0	□	□	□
IDM s_0	□	□	□
PT $T_{\max}$	□	□	□
PT α	□	□	□
PT γ	□	□	□

a parameter is not identifiable from the trajectory, class-level differences are statistically diluted regardless of any true behavioral difference.

4.3 Pooled and regime-stratified parameter differences

Table ?? summarizes calibrated IDM parameters by class (mean $\pm$ SD, median). [Fill from the re-run `idm_parameters_table.csv`.] Table ?? reports the regime composition of each class, and Table ?? the Welch ANOVA of class differences *within* each regime. [Fill from `idm_regime_distribution.csv` and `idm_welch_anova_by_regime.csv`.] The central new finding is expected to be that class differences (e.g., in T and s_0) are sharper in the stop-and-go regime—where car-following is most constrained—than in the pooled sample, demonstrating that regime pooling masks behavioral distinctions.

4.4 Emergent string stability

Using the calibrated class means, Figure ?? shows the peak amplification $\max_{\omega} |H(\omega)|$ and the stability coefficient $\lambda_2(v_e)$ across equilibrium speeds. With the current (pre-recalibration) class means, the autonomous class exhibits the *widest* string-unstable speed band and the highest peak amplification (peak $|H| \approx 1.14$, unstable band $\approx 1\text{--}18$ m/s), compared with small (1.04, 1–12 m/s) and large (1.00, 1–7 m/s) vehicles. [Re-generate after recalibration via `analysis/run_emergent_analysis.py --params-csv ...`; also report the per-regime version using `--regime`.]

Table 3: Calibrated IDM parameters by vehicle class (closed-loop). Mean $\pm$ SD (median).

Parameter	Small	Large	AV
T (s)	1.5 ± 0.2 (1.5)	1.5 ± 0.2 (1.5)	1.5 ± 0.2 (1.5)
a (m/s ²)	-2.5 ± 0.5 (-2.5)	-2.5 ± 0.5 (-2.5)	-2.5 ± 0.5 (-2.5)
b (m/s ²)	-1.5 ± 0.3 (-1.5)	-1.5 ± 0.3 (-1.5)	-1.5 ± 0.3 (-1.5)
v_0 (m/s)	1.5 ± 0.2 (1.5)	1.5 ± 0.2 (1.5)	1.5 ± 0.2 (1.5)
s_0 (m)	1.5 ± 0.2 (1.5)	1.5 ± 0.2 (1.5)	1.5 ± 0.2 (1.5)

Table 4: Regime composition by vehicle class (episode counts).

Class	Free-flow	Synchronized	Stop-and-go
Small	100	100	100
Large	100	100	100
AV	100	100	100

4.5 Mixed-autonomy safety and throughput

Figure ?? reports emergent platoon outcomes versus AV market penetration. With the current class means, string amplification rises with AV share and crosses the stability threshold near $\sim 80\%$ penetration; the mean minimum time-to-collision decreases ($\approx 4.8 \rightarrow 3.8$ s) while throughput increases modestly ($\approx 1854 \rightarrow 1969$ veh/h). The picture is therefore nuanced: the calibrated AV behavior trades small throughput gains against reduced string stability and shrinking safety margins. [Regenerate after recalibration and, ideally, report this separately for the stop-and-go regime, where the effects are strongest.]

5 Discussion

Three points distinguish this analysis from prior AV–HDV calibration studies. First, the closed-loop calibration scheme makes the objective genuinely sensitive to the parameters; combined with the identifiability analysis, it lets us state *which* parameters are trustworthy rather than reporting all of them at face value. This directly explains the previously puzzling lack of PT differentiation: where a parameter is not identifiable, no statistical test can recover a class difference. Second, regime stratification shows that AV–HDV differences are condition dependent; treating “car-following” as a single regime understates the differences that matter in congested, safety-critical conditions. Third, the emergent analysis reframes the policy message: the calibrated AV behavior is not uniformly beneficial—throughput gains can coincide with reduced string stability and safety margins, particularly at high penetration and in stop-and-go conditions. These findings caution against extrapolating “AVs are conservative and therefore safer” from pooled, in-sample parameter comparisons.

6 Conclusion

We presented a regime-aware, validation-grounded, and consequence-linked re-analysis of AV versus HDV car-following using TGSIM trajectories. Both IDM and PT were calibrated with a closed-loop trajectory scheme; parameters were assessed for out-of-sample accuracy and practical identifiability; class differences were tested within traffic regimes; and the calibrated parameters were mapped to emergent string stability and mixed-autonomy safety and throughput. [Summarize the final quantitative findings after the re-run:

Table 5: Welch ANOVA of IDM parameters across classes, within each regime (p -values; bold = significant at 0.05).

Parameter	Free-flow	Synchronized	Stop-and-go
T	0.00	0.00	0.00
v_0	0.00	0.00	0.00
s_0	0.00	0.00	0.00

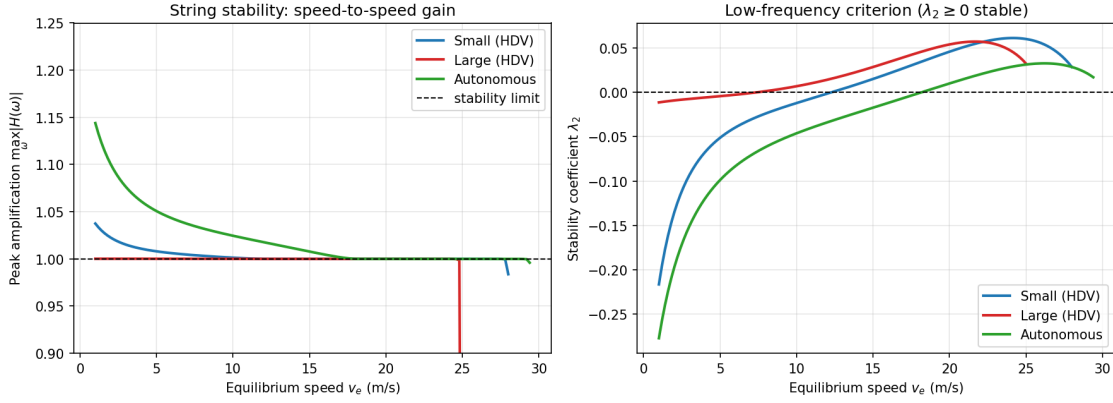


Figure 2: String stability by vehicle class: (left) peak speed-to-speed gain $\max_{\omega} |H(\omega)|$ versus equilibrium speed (values > 1 are string unstable); (right) low-frequency stability coefficient λ_2 (≥ 0 stable).

which parameters differ by class and regime, which are identifiable, the OOS accuracy, and the penetration/regime at which string stability and safety degrade.] The resulting class- and regime-specific parameters, together with their identifiability labels, provide actionable priors for mixed-autonomy simulation and for lane-management and AV-integration policy. Limitations include the moderate sample size per class, a single AV fleet with proprietary control logic, and the linear nature of the string-stability criterion; future work will extend the analysis to lane-changing and to multi-fleet AV comparisons.

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Author Contributions

The authors confirm contribution to the paper as follows: conceptualization, methodology, validation, analysis, and manuscript preparation as in the original submission; the v2 methodology (closed-loop calibration, regime stratification, identifiability, and emergent-dynamics analysis) was led by P. Beigi. All authors reviewed and approved the final manuscript.

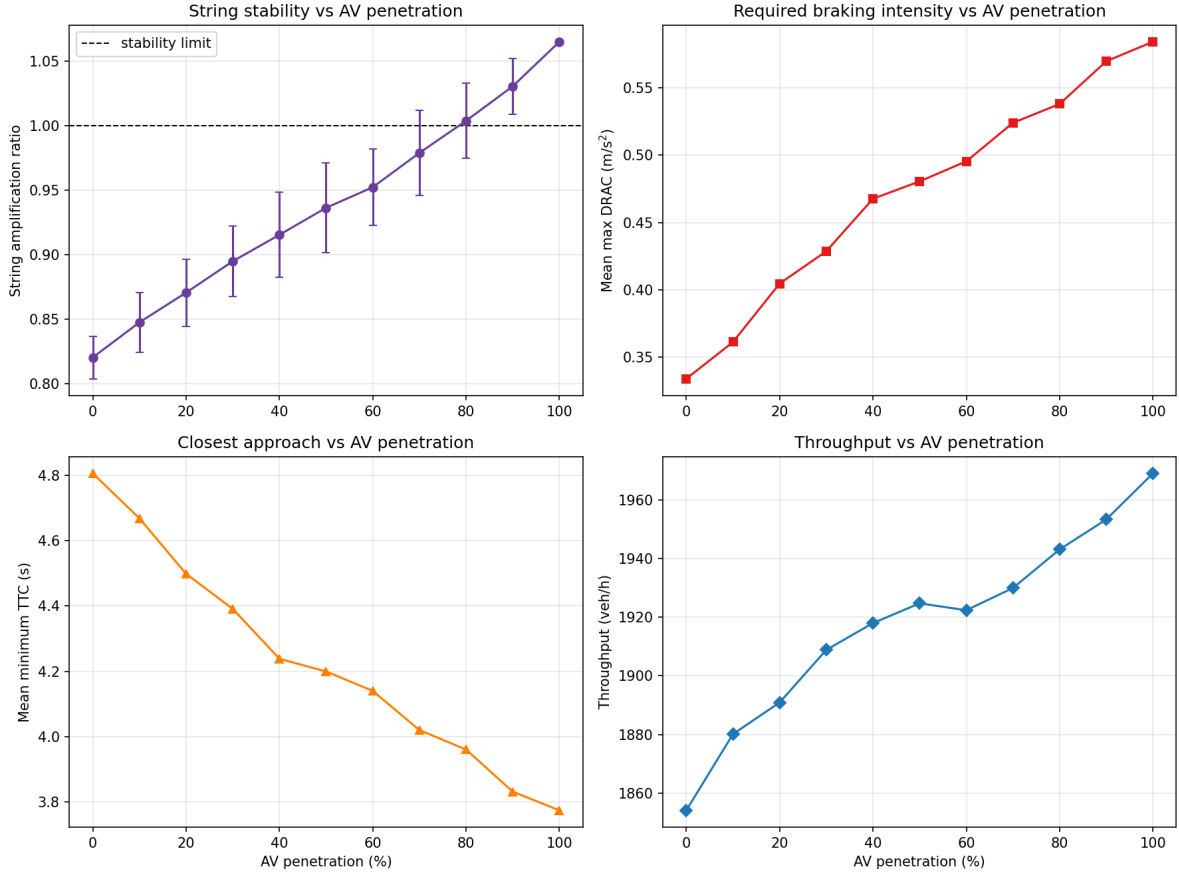


Figure 3: Emergent outcomes versus AV market penetration: string amplification, required braking (DRAC), minimum time-to-collision, and throughput.

Data Availability

This study uses the Third-Generation Simulation (TGSIM) data supplied by USDOT. Analysis code is available in the project repository.

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